UQA Technical Overview

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"Logic pervades the world: the limits of the world are also its limits."
— Ludwig Wittgenstein

This document provides a detailed technical exposition of the Unified Query Algebra, connecting the formal theory developed across three research papers to the concrete implementation in the UQA codebase.

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1. Posting Lists as Universal Abstraction

1.1 The Core Thesis

UQA's foundational claim is that a single data structure — the **posting list** — can serve as the universal abstraction for result sets across relational, text retrieval, vector search, and graph query paradigms.

Definition. A posting list L is an ordered sequence of entries:

$$L = [(id_1, payload_1),\ (id_2, payload_2),\ \ldots,\ (id_n, payload_n)] \tag{1}$$

where $id_i < id_j$ for all $i < j$ (sorted by document ID), and each payload carries positions, scores, and field values.

A bijection $PL : 2^{\mathcal{D}} \to \mathcal{L}$ maps document sets to posting lists and back, allowing set-theoretic reasoning to carry over directly.

1.2 Boolean Algebra

The structure $(\mathcal{L},\ \cup,\ \cap,\ \overline{\cdot},\ \emptyset,\ \mathcal{D})$ forms a **complete Boolean algebra** satisfying commutativity, associativity, distributivity, identity, and complement laws.

Law	Equation
Commutativity	$A \cup B = B \cup A$, $A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$
Distributivity	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \emptyset = A$, $A \cap \mathcal{D} = A$
Complement	$A \cup \overline{A} = \mathcal{D}$, $A \cap \overline{A} = \emptyset$

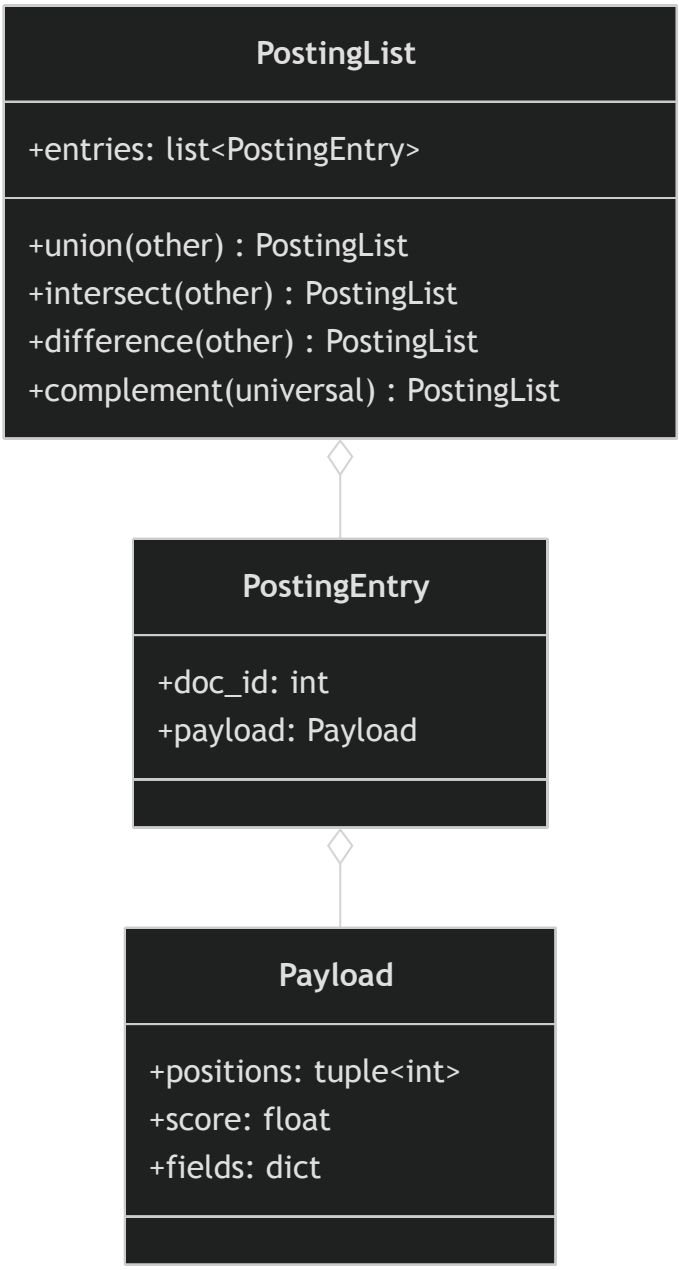
This algebraic closure guarantees that any combination of AND, OR, and NOT across paradigms is well-defined, and query optimization can exploit lattice-theoretic rewrite rules.

1.3 Implementation

The `PostingList` class in `uqa/core/posting_list.py` implements the algebra directly:

```
PostingList
  entries: list[PostingEntry]       # sorted by doc_id
  union(other) -> PostingList       # two-pointer merge, all doc_ids
  intersect(other) -> PostingList   # two-pointer merge, shared doc_ids
  difference(other) -> PostingList  # entries in self but not other
  complement(universal) -> PostingList
```

Operator overloads map to algebra: `|` for union, `&` for intersect, `-` for difference. On merge, payloads are combined: positions are unioned, scores are summed, field dictionaries are merged. The `GeneralizedPostingList` extends this to join results where entries hold tuples of document IDs.



2. Cross-Paradigm Operators

2.1 Primitive Operators

Each paradigm maps into the posting list space through a primitive operator:

Operator	Definition	Paradigm	Implementation
$T(t)$	$PL(\{d \in \mathcal{D} \mid t \in term(d, f)\})$	Text retrieval	`TermOperator`
$V_\theta(q)$	$PL(\{d \in \mathcal{D} \mid sim(vec(d, f), q) \geq \theta\})$	Vector search	`VectorSimilarityOperator`
$KNN_k(q)$	$PL(D_k)$ where $\|D_k\| = k$, ranked by similarity	Vector search	`KNNOperator`
$Filter_{f,v}(L)$	$L \cap PL(\{d \in \mathcal{D} \mid d.f = v\})$	Relational	`FilterOperator`
$Score_q(L)$	$(L,\ [s_1, \ldots, s_{\|L\|}])$	Scoring	`ScoreOperator`

Because every operator produces a posting list, they compose freely:

$$Hybrid_{t,q,\theta} = T(t) \cap V_\theta(q) \tag{2}$$

2.2 Operator Monoid

All operators form a monoid under composition. The abstract base class `Operator` in `uqa/operators/base.py` exposes:

- `execute(context: ExecutionContext) -> PostingList` — the universal contract
- `compose(other) -> ComposedOperator` — sequential composition (monoid product)
- `cost_estimate(stats) -> float` — for optimizer decisions

The `ExecutionContext` bundles all storage backends: `DocumentStore`, `InvertedIndex`, `HNSWIndex`, `GraphStore`, `BlockMaxIndex`, `IndexManager`, and `ParallelExecutor`.

2.3 Boolean Operators

Operator	Theory	Implementation
Union $A_1 \cup A_2 \cup \ldots \cup A_n$	Boolean OR	`UnionOperator`
Intersection $A_1 \cap A_2 \cap \ldots \cap A_n$	Boolean AND	`IntersectOperator`
Complement $\mathcal{D} \setminus A$	Boolean NOT	`ComplementOperator`

`UnionOperator` and `IntersectOperator` support parallel execution of their operands via `ParallelExecutor` (ThreadPoolExecutor), with results reduced through `PostingList.union` or `PostingList.intersect`.

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3. Scoring: From BM25 to Bayesian BM25

3.1 Classical BM25

Robertson and Zaragoza derived BM25 from a probabilistic model of term occurrence. The score for a document d given query term t is:

$$BM25(d,t) = IDF(t) \cdot \frac{tf(t,d) \cdot (k_1 + 1)}{tf(t,d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{avgdl}\right)} \tag{3}$$

where $IDF(t) = \ln\left(\frac{N - df(t) + 0.5}{df(t) + 0.5} + 1\right)$ is the Robertson-Sparck Jones inverse document frequency.

Implementation. `BM25Scorer` in `uqa/scoring/bm25.py` implements a numerically stable reformulation:

$$score = w - \frac{w}{1 + f \cdot inv_norm} \tag{4}$$

where $w = boost \cdot IDF$ and $inv_norm = 1/(k_1 \cdot ((1-b) + b \cdot dl/avgdl))$.

The upper bound $ub(t) = boost \cdot IDF(t)$ supports safe document pruning in WAND/BMW.

3.2 The Probabilistic Relevance Gap

BM25 scores are not probabilities. They lie in $[0, +\infty)$ with query-dependent magnitudes, making them incompatible with other ranking signals and violating Robertson's own Probability Ranking Principle. For nearly five decades, this gap persisted.

3.3 Bayesian BM25

Bayesian BM25 closes the gap by applying Bayesian inference with a sigmoid likelihood model to transform BM25 scores into calibrated probabilities $P(R = 1 \mid s) \in [0, 1]$.

Sigmoid Posterior. Under a symmetric likelihood assumption with log-linear likelihood ratio:

$$P(R = 1 \mid s) = \sigma(\alpha(s - \beta)) \tag{5}$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the logistic sigmoid.

Three-Term Decomposition. The full posterior decomposes in log-odds space (additive decomposition of Bayes' theorem):

$$\text{logit}(P_{posterior}) = \underbrace{\text{logit}(L(s))}_{\text{likelihood}} + \underbrace{\text{logit}(b_r(f, \hat{n}))}_{\text{doc-dependent prior}} + \underbrace{\text{logit}(p)}_{\text{base rate}} \tag{6}$$

where $L(s) = \sigma(\alpha(s - \beta))$ is the normalized likelihood ratio, $b_r(f, \hat{n})$ is a composite prior incorporating term frequency rarity and document length, and p is the corpus-level base rate. This reduces expected calibration error by 68-77% without requiring relevance labels.

Implementation. `BayesianBM25Scorer` in `uqa/scoring/bayesian_bm25.py` delegates to the `bayesian-bm25` package, which implements the full three-term posterior with progressive hyperparameter estimation.

3.4 Vector Similarity

Cosine similarity $\cos\theta \in [-1, 1]$ is linearly calibrated to probability:

$$P_{vec} = \frac{1 + \cos\theta}{2} \tag{7}$$

This mapping is exact for the case where vector similarity represents angular proximity. `VectorScorer` in `uqa/scoring/vector.py` delegates to `bayesian-bm25`'s `cosine_to_probability()` for numerical stability.

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4. Fusion: Combining Cross-Paradigm Signals

4.1 The Conjunction Shrinkage Problem

Given n independent calibrated relevance signals, the naive probabilistic conjunction:

$$P_{AND} = \prod_{i=1}^{n} P_i \tag{8}$$

suffers from **conjunction shrinkage**: as n increases, $P_{AND} \to 0$ even when all signals report high relevance. This violates the fundamental intuition that agreement among independent sources should increase confidence.

4.2 Log-Odds Conjunction

The log-odds conjunction resolves this by working in the natural parameter space of the Bernoulli distribution:

$$P_{final} = \sigma\left(\frac{1}{n^{1-\alpha}} \sum_{i=1}^{n} \text{logit}(P_i)\right) \tag{9}$$

where $\text{logit}(p) = \log \frac{p}{1-p}$ and α controls confidence scaling.

Key properties:

Property	Guarantee
Scale neutrality	All $P_i = p \implies P_{final} = p$ regardless of n
Sign preservation	$\text{sgn}(\ell_{adjusted}) = \text{sgn}(\bar{\ell})$
Irrelevance preservation	All $P_i < 0.5 \implies P_{final} < 0.5$ for all n
Agreement amplification	All $P_i > 0.5 \implies P_{final} > \sigma(\bar{\ell})$ for $n \geq 2$

Setting $\alpha = 0.5$ yields the classical $\sqrt{n}$ confidence scaling law, where per-signal weight is $w_i = 1/\sqrt{n}$.

This is mathematically equivalent to the **normalized Logarithmic Opinion Pool** (Log-OP), also known as Hinton's **Product of Experts** (PoE, 2002):

$$\text{logit}(P_{Log-OP}) = \sum_{i=1}^{n} \frac{1}{n} \text{logit}(P_i) = \bar{\ell} \tag{10}$$

Numerical comparison ($n = 2$, $\alpha = 0.5$):

P_{text}	P_{vec}	Product	Log-Odds	Interpretation
0.9	0.9	0.81	0.96	Strong agreement amplified
0.7	0.7	0.49	0.77	Moderate agreement preserved
0.7	0.3	0.21	0.50	Exact neutrality (logits cancel)
0.3	0.3	0.09	0.23	Irrelevance preserved

4.3 Probabilistic Boolean Operations

For explicit Boolean query semantics:

Operation	Formula	Log-space
AND	$P = \prod P_i$	$\log P = \sum \log P_i$
OR	$P = 1 - \prod(1 - P_i)$	$\log(1 - P) = \sum \log(1 - P_i)$
NOT	$P = 1 - P_{signal}$	

All computed in log-space for numerical stability.

4.4 Implementation

Function	Class	File
`fuse_log_odds(...)`	`LogOddsFusion`	`uqa/fusion/log_odds.py`
`fuse_prob_and(...)`	`ProbabilisticBoolean`	`uqa/fusion/boolean.py`
`fuse_prob_or(...)`	`ProbabilisticBoolean`	`uqa/fusion/boolean.py`
`fuse_prob_not(...)`	`ProbabilisticBoolean`	`uqa/fusion/boolean.py`

In SQL:

```sql
SELECT title, _score FROM papers
WHERE fuse_log_odds(
    text_match(title, 'attention'),
    knn_match(embedding, ARRAY[0.1, 0.2, ...], 5)
) ORDER BY _score DESC;
```

5. Graph Extension

5.1 Graph-Posting List Isomorphism

The second paper extends the framework to graph data by establishing an isomorphism between graph posting lists and standard posting lists.

Graph Posting List. A graph posting list $L_G = [(id_1, G_1), \ldots, (id_n, G_n)]$ maps to standard posting lists via:

$$\Phi(L_G) = PL\left(\bigcup_{i=1}^{n} \phi_{G\to D}(G_i)\right) \tag{11}$$

This isomorphism preserves Boolean operations:

$$\Phi(L_G^1 \cup_G L_G^2) = \Phi(L_G^1) \cup \Phi(L_G^2) \tag{12}$$

$$\Phi(L_G^1 \cap_G L_G^2) = \Phi(L_G^1) \cap \Phi(L_G^2) \tag{13}$$

Implementation. `GraphPostingList` in `uqa/graph/posting_list.py` extends `PostingList`, providing `to_posting_list()` (Φ) and `from_posting_list()` (Φ^{-1}). Subgraph structure is encoded in `Payload.fields` as frozen sets of vertex/edge IDs.

5.2 Graph Operators

Operator	Definition	Complexity	Implementation
$Traverse_{v,l,k}$	BFS from v, edges matching l, up to k hops	$O(\sum_{i=1}^{k} d^i)$	`TraverseOperator`
$GMatch_P$	Subgraph isomorphism via backtracking	$O(\lVert V_G \rVert^{\lVert V_P \rVert})$ (NP-complete)	`PatternMatchOperator`
RPQ_R	Regular path query via NFA simulation	$O(\lVert V_G \rVert^2 \cdot \lVert R \rVert)$	`RegularPathQueryOperator`
$VertexAgg_{p,f}$	Aggregate property p with function f	$O(\lVert L_G \rVert)$	`VertexAggregationOperator`

5.3 Subgraph Isomorphism: Complexity and Mitigations

Pattern matching via subgraph isomorphism ($GMatch_P$) has worst-case complexity $O(\|V_G\|^{\|V_P\|})$ and is NP-complete. This is an inherent property of the problem itself — any system supporting general subgraph pattern matching faces the same theoretical bound. The key question is how to keep the **effective** search space tractable.

Naive backtracking iterates all $\|V_G\|$ vertices for each of the $\|V_P\|$ pattern variables in a fixed order, checking edge constraints only after all variables are assigned. For a pattern with k variables on a graph with n vertices, this explores up to n^k complete assignments before discovering that most are invalid.

UQA applies three CSP (Constraint Satisfaction Problem) techniques that reduce the effective search space by orders of magnitude:

1. Candidate Pre-computation with Arc Consistency.
Before backtracking begins, each pattern variable's candidate set is computed by evaluating vertex constraints once against all graph vertices. Then an arc consistency fixpoint loop propagates edge constraints: if variable A must connect to variable B via a labeled edge, any candidate for A that has no such outgoing edge to any candidate of B is eliminated, and vice versa. This propagation repeats until no further reduction occurs.

$$C_v^{(0)} = \{u \in V_G \mid u \text{ satisfies vertex constraints of } v\} \tag{14}$$

$$C_v^{(i+1)} = \{u \in C_v^{(i)} \mid \forall (v, w, l) \in E_P,\ \exists u' \in C_w^{(i)},\ (u, u', l) \in E_G\} \tag{15}$$

This is computed to fixpoint ($C^{(i+1)} = C^{(i)}$), typically converging in 1-2 iterations. The reduction can be dramatic: on a graph with 1M vertices where only 1K have label `Person`, the candidate set shrinks by 1000x before any backtracking occurs.

2. MRV (Minimum Remaining Values) Variable Ordering.
At each recursion step, the algorithm selects the unassigned variable with the fewest remaining candidates — the "fail-first" heuristic. If one variable has 3 candidates and another has 10,000, trying the 3-candidate variable first means at most 3 branches explored before pruning occurs, rather than 10,000.

3. Incremental Edge Validation.
Instead of deferring all edge checks to the end (after all k variables are assigned), each variable binding immediately validates all edge constraints connecting the newly bound variable to previously bound variables. Invalid partial assignments are pruned at depth 2 rather than depth k.

The combined effect:

Optimization	Reduction Factor	Mechanism
Candidate pre-computation	$\|V_G\| \to \|C_v\|$ per variable	Domain initialization
Arc consistency	$\|C_v\| \to \|C'_v\| \leq \|C_v\|$	Edge-based propagation
MRV ordering	Exponential pruning	Fail-first branching
Incremental edge validation	Early cutoff at depth $d \ll k$	Forward checking

In practice, Cypher patterns typically contain 3-7 nodes with label constraints, reducing the search to a small fraction of the graph. The theoretical $O(n^k)$ worst case manifests only for unlabeled patterns on dense graphs with no distinguishing constraints — a scenario rarely encountered in real workloads.

Implementation. `PatternMatchOperator` in `uqa/graph/operators.py` implements all three techniques. `CypherCompiler` in `uqa/graph/cypher/compiler.py` applies label-based candidate filtering via `vertices_by_label()` and incremental vertex/edge validation during path expansion. Variable-length relationship expansion uses BFS with `deque` for O(1) frontier access.

5.4 Regular Path Expressions

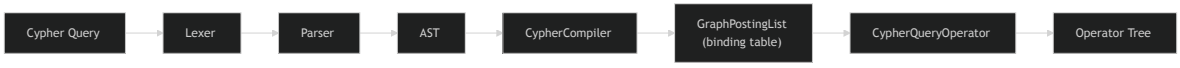
Regular path queries use an algebraic expression language over edge labels:

$$R ::= l \;\mid\; R_1 \cdot R_2 \;\mid\; R_1 | R_2 \;\mid\; R^* \tag{16}$$

The `RegularPathExpr` hierarchy (`Label`, `Concat`, `Alternation`, `KleeneStar`) in `uqa/graph/pattern.py` represents these expressions. The `RegularPathQueryOperator` uses Thompson's construction to build an NFA from the expression, then simulates it on the graph.

5.5 openCypher Integration

The `CypherCompiler` in `uqa/graph/cypher/` compiles openCypher queries into graph posting list operations:



Every Cypher clause transforms a **binding table** (represented as a `GraphPostingList`). The `CypherQueryOperator` integrates into the standard operator tree — it is not a bypass but a full posting-list citizen, visible in `EXPLAIN` output.

SQL integration via FROM-clause table functions:

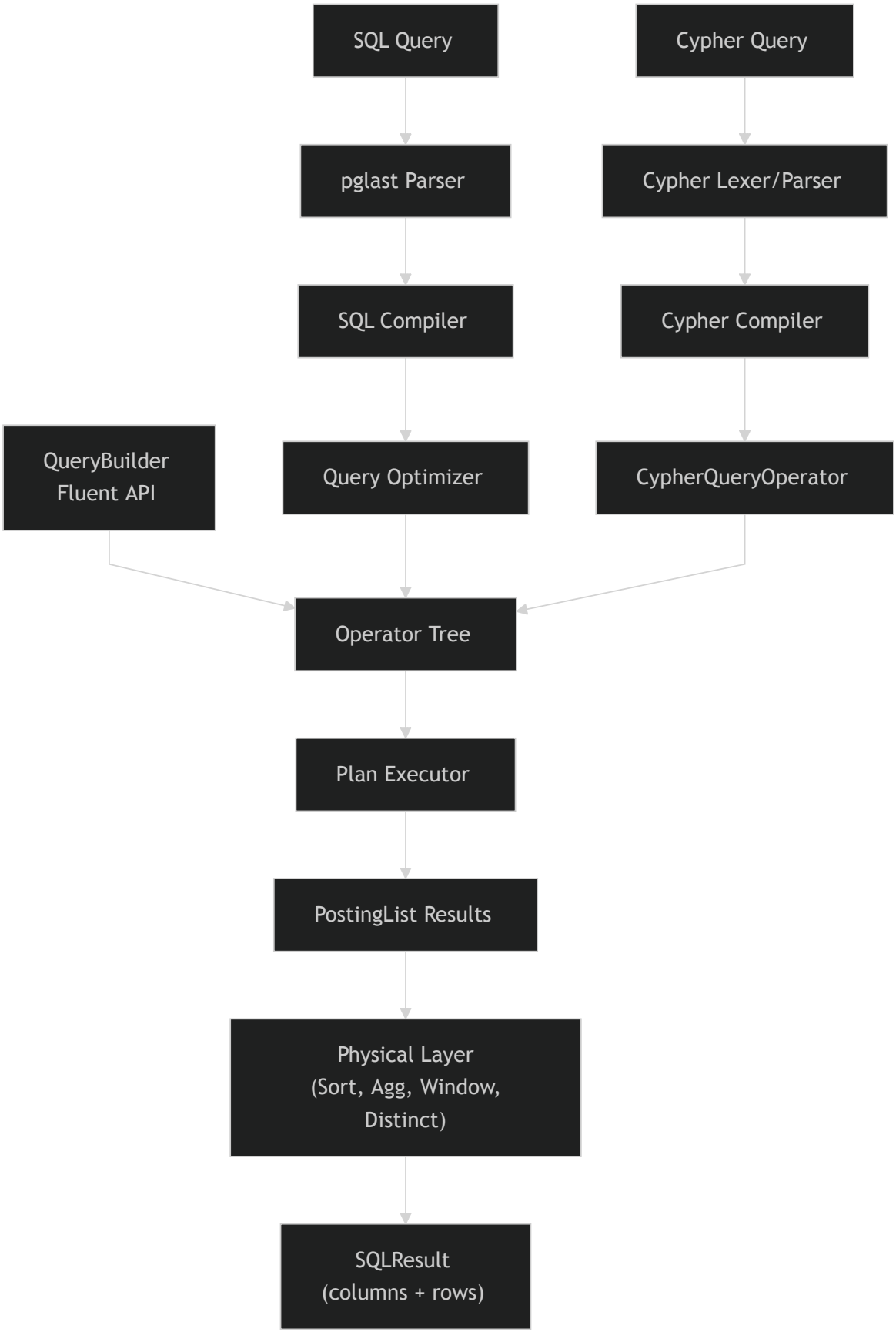
```
SELECT * FROM cypher('social', $$
    MATCH (p:Person)-[:KNOWS]->(f:Person)
    WHERE p.age > 25
    RETURN p.name, f.name
$$) AS (person agtype, friend agtype);
```

---

6. System Architecture

6.1 Query Pipeline

All queries — regardless of entry point (SQL, Cypher, Fluent API) — produce intermediate and final results as posting lists:



6.2 SQL Compiler Pipeline

The SQL compiler in `uqa/sql/compiler.py` follows a staged pipeline:

1. **Parse.** pglast (PostgreSQL parser) produces an AST.

2. **Route.** Statement type dispatch (SELECT, INSERT, UPDATE, DELETE, DDL, utility).
3. **Compile SELECT:**
    - Materialize CTEs as temporary tables
    - Resolve FROM clause to (Table, source_op)
    - Create `ExecutionContext` from table storage
    - Process WHERE clause into operator tree
    - Optimize via `QueryOptimizer`
    - Execute operator tree to `PostingList`
    - Apply relational layer (GROUP BY, HAVING, ORDER BY, LIMIT, DISTINCT) via physical operators

6.3 Physical Execution

The physical layer in `uqa/execution/` uses the **Volcano iterator model**: each `PhysicalOperator` implements `open() / next() / close()`, pulling batches from children on demand (pipelined execution).

Operator	Type	Strategy
`FilterOp`	Streaming	Predicate evaluation per batch
`ExprFilterOp`	Streaming	Complex expressions via `ExprEvaluator`
`ProjectOp`	Streaming	Column selection
`SortOp`	Blocking	External merge sort (spills to Arrow IPC)
`HashAggOp`	Blocking	Grace hash partitioning (spills to disk)
`DistinctOp`	Blocking	Hash partition dedup
`LimitOp`	Streaming	LIMIT/OFFSET
`WindowOp`	Blocking	Window function evaluation

6.4 Storage Architecture

Each SQL table maintains isolated storage instances:

```
Engine
+-- _tables: dict[str, Table]
|   +-- Table
|       +-- DocumentStore         (typed columnar storage)
|       +-- InvertedIndex         (term -> posting list per field)
|       +-- HNSWIndex             (optional, per VECTOR column)
|       +-- ColumnStats           (per-column statistics for optimizer)
+-- _named_graphs: dict[str, GraphStore]
|   +-- GraphStore                (adjacency list + property maps)
+-- _foreign_servers: dict[str, ForeignServer]
+-- _foreign_tables: dict[str, ForeignTable]
+-- _fdw_handlers: dict[str, FDWHandler]
```

```
+-- _catalog: Catalog             (SQLite, write-through persistence)
+-- _transaction: Transaction     (BEGIN/COMMIT/ROLLBACK/SAVEPOINT)
```

When `db_path` is provided, all mutations are write-through to SQLite. On startup, `_load_from_catalog()` restores the full state.

6.5 Query Optimizer

The cost-based optimizer in `uqa/planner/` applies:

- **DPccp join order optimization** (Moerkotte & Neumann, 2006) — $O(3^n)$ dynamic programming over connected subgraph complement pairs for optimal bushy join trees; uses integer bitmask DP table, bytearray connectivity lookup, incremental connected subgraph enumeration, and canonical submask enumeration for efficient inner-loop execution
- Filter pushdown into intersections
- Vector threshold merge (same query vector)
- Intersect operand reordering by cardinality (cheapest first)
- Fusion signal reordering by cost
- B-tree index scan substitution
- FDW predicate pushdown (comparison, IN, LIKE, ILIKE, BETWEEN)

Cardinality estimation uses equi-depth histograms and Most Common Values (MCV), with specialized estimators for text, vector, graph, and fusion operators.

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7. Theory-to-Code Mapping

The table below maps each formal definition and theorem from the three papers to its concrete implementation:

Paper 1: Unified Mathematical Framework

Theory	Implementation
Posting List (Definition 2.1.1)	`PostingList` in `uqa/core/posting_list.py`
Boolean Algebra (Theorem 2.1.2)	`union()`, `intersect()`, `difference()`, `complement()`
Generalized Posting List (Definition 4.1.2)	`GeneralizedPostingList` in `uqa/core/posting_list.py`
Operator Monoid (Theorem 3.2.3)	`Operator` ABC + `compose()` in `uqa/operators/base.py`
Term Operator $T(t)$ (Definition 3.1.1)	`TermOperator` in `uqa/operators/primitive.py`
Vector Operators V_θ, KNN_k (Definitions 3.1.2, 3.1.3)	`VectorSimilarityOperator`, `KNNOperator`
Filter Operator (Definition 3.1.4)	`FilterOperator` in `uqa/operators/primitive.py`
Score Operator (Definition 3.1.5)	`ScoreOperator` in `uqa/operators/primitive.py`
Hierarchical Documents (Definition 5.2.1)	`HierarchicalDocument` in `uqa/core/hierarchical.py`
BM25 (Definition 3.2.1)	`BM25Scorer` in `uqa/scoring/bm25.py`
UnionOperator	`UnionOperator` in `uqa/operators/boolean.py`
IntersectOperator	`IntersectOperator` in `uqa/operators/boolean.py`
ComplementOperator	`ComplementOperator` in `uqa/operators/boolean.py`

Paper 2: Graph Extension

Theory	Implementation
Graph-Posting List Isomorphism (Theorem 1.1.6)	`GraphPostingList` in `uqa/graph/posting_list.py`
Traversal Operator (Definition 2.2.1)	`TraverseOperator` in `uqa/graph/operators.py`
Pattern Match (Definition 5.2.2)	`PatternMatchOperator` in `uqa/graph/operators.py`
Regular Path Query (Definition 5.1.2)	`RegularPathQueryOperator` in `uqa/graph/operators.py`
Vertex Aggregation (Definition 2.2.3)	`VertexAggregationOperator` in `uqa/graph/operators.py`
Graph Pattern Expression (Definition 5.2.1)	`GraphPattern` in `uqa/graph/pattern.py`
Regular Path Expression (Definition 5.1.1)	`RegularPathExpr` hierarchy in `uqa/graph/pattern.py`
openCypher (cross-paradigm integration)	`CypherCompiler` in `uqa/graph/cypher/`

Paper 3: Bayesian BM25

Theory	Implementation
Sigmoid Posterior (Theorem 4.1.1)	`BayesianBM25Scorer.score()` in `uqa/scoring/bayesian_bm25.py`
Three-Term Decomposition (Theorem 4.4.2)	Two successive Bayes updates in `bayesian-bm25` package
Vector Calibration (Definition 7.1.2)	`VectorScorer.similarity_to_probability()` in `uqa/scoring/vector.py`
Probabilistic AND (Theorem 5.1.1)	`ProbabilisticBoolean.prob_and()` in `uqa/fusion/boolean.py`
Probabilistic OR (Theorem 5.2.1)	`ProbabilisticBoolean.prob_or()` in `uqa/fusion/boolean.py`
WAND Upper Bound (Theorem 6.1.2)	`BayesianBM25Scorer.upper_bound()`

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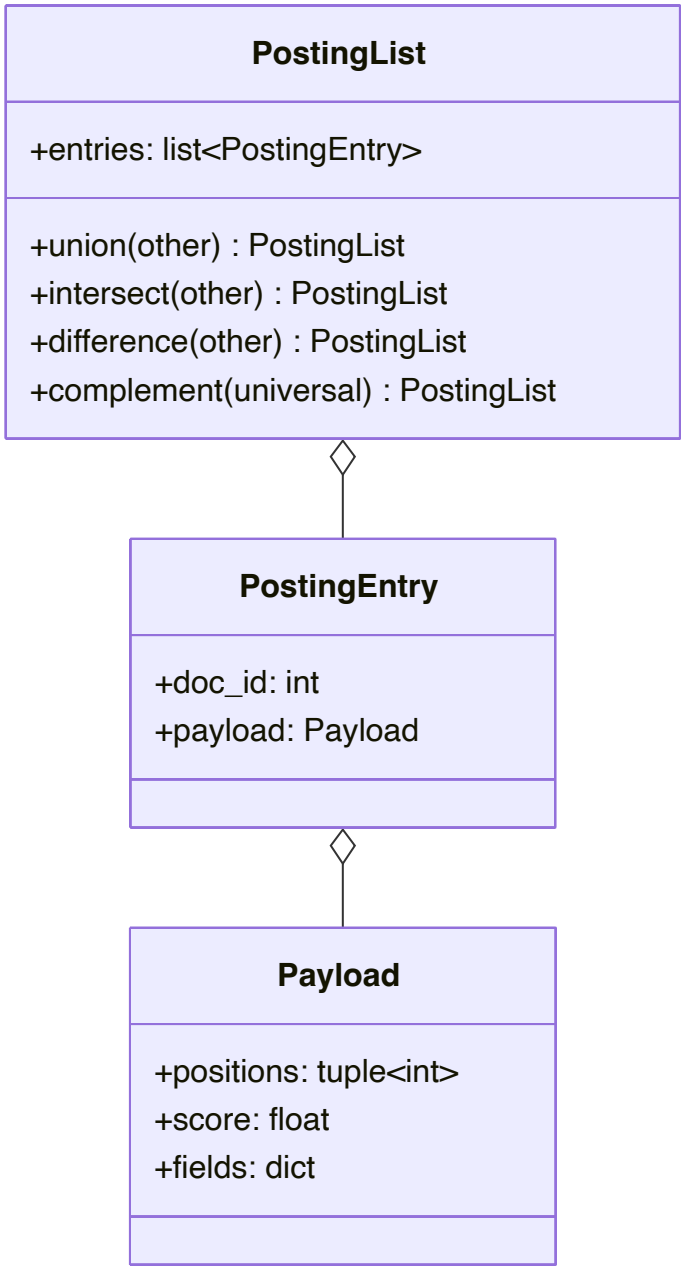
8. Cross-Paradigm Composition

The key invariant — **every query produces a PostingList** — enables seamless cross-paradigm composition:



In SQL, this is a single query:

```
SELECT title, _score FROM papers
WHERE fuse_log_odds(
    text_match(title, 'attention'),
    knn_match(embedding, ARRAY[0.1, 0.2, ...], 5),
    traverse_match(1, 'cited_by', 2)
) AND year >= 2020
ORDER BY _score DESC;
```

The query optimizer ensures efficient execution: each paradigm's operator executes independently (potentially in parallel), their posting lists are intersected via the fusion operator, and the relational filter is applied on the fused result.

References

1. Jeong, J. (2023). A Unified Mathematical Framework for Query Algebras Across Heterogeneous Data Paradigms.
2. Jeong, J. (2024). Extending the Unified Mathematical Framework to Support Graph Data Structures.
3. Jeong, J. (2026). Bayesian BM25: A Probabilistic Framework for Hybrid Text and Vector Search.